

IGCSE Computer Science 0478 — Key Facts & Reference

Cambridge IGCSE 0478 (2023–2026). CS has no formulas — this sheet is the equivalents: number systems, logic, units, and the rules you must recall.

NUMBER SYSTEMS

Binary to denary	place values 128 64 32 16 8 4 2 1 — sum the 1s
Binary to hex	groups of 4 bits from the right: 1010 = A, 1111 = F
Hex to denary	$\times 16$ per left shift: $2F = 2 \times 16 + 15 = 47$
Two's complement	flip bits, add 1 (negative 8-bit range: -128 to +127)
Binary shifts	left $\times 2$ per shift (overflow risk); right $\div 2$ per shift (loss of precision)
Character sets	ASCII: 7-bit (128 chars); Unicode: covers all languages

SIZES & UNITS

Bit to byte	8 bits = 1 byte
File sizes	1 KB = 1,000 B; 1 MB = 1,000 KB; 1 GB = 1,000 MB (exam convention)
Calculation	file size = bits per sample $\times$ samples/s $\times$ seconds $\times$ channels

LOGIC GATES & BOOLEAN

NOT	output = opposite of input
AND	output 1 only when BOTH inputs are 1
OR	output 1 when AT LEAST one input is 1
NAND / NOR	AND/OR followed by NOT
XOR	output 1 when inputs are DIFFERENT

Always build the truth table (2^n rows for n inputs) — it earns the marks even if the circuit

PSEUDOCODE & ALGORITHMS

Assignment	$x \leftarrow 5$ ($! = \text{not} =$)
Selection	IF ... THEN ... ELSE ... ENDIF; CASE ... ENDCASE
Loops	FOR ... NEXT; WHILE ... ENDWHILE; REPEAT ... UNTIL
Arrays	1-indexed in exam pseudocode: A[1..n]
Linear search	check each item in turn — $O(n)$
Bubble sort	repeated adjacent swaps until no swaps — $O(n^2)$

Trace tables: track EVERY variable, one line per iteration. Don't skip rows — examiners award marks per correct row.

DATA REPRESENTATION

Sound	sample rate (Hz) $\times$ sample resolution (bits) $\times$ time $\times$ channels
Images	resolution (w $\times$ h) $\times$ colour depth (bits) = file size in bits
Compression	lossless (RLE, dictionary) — original restored; lossy — smaller, quality lost

NETWORKS & SECURITY (recall facts)

Protocols	HTTP(S), FTP, SMTP (send), POP/IMAP (receive), UDP vs TCP
Encryption	symmetric = same key both ways; asymmetric = public + private key pair
Cyber threats	malware, phishing, brute force, DoS, interception — know a defence each